

RTS26 – INFINITY AND CYCLES:

AN RTS PERSPECTIVE ON CONTEMPORARY COSMOLOGY

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ABSTRACT

Contemporary cosmology faces open questions: is the universe finite or infinite? Is energy globally conserved? Are Penrose's cycles real? This episode does not offer definitive answers, but a reinterpretation of these questions through the lens of RTS. We show that the question "finite or infinite" is wrongly formulated: our universe is a stable slice of a larger Network, which may be infinite in possibilities but finite in manifestation. Cycles are phase transitions of the Network, and energy is a local property of couplings, not a global constant.

Keywords: infinity, cycles, RTS, Penrose, Greene, energy, phase transition, Primary Network.

1. INTRODUCTION – WHAT COSMOLOGY ASKS

Contemporary cosmology asks:

- Is the universe infinite?
- How does energy behave at cosmic scales?
- Are Penrose's cycles real?

These questions are legitimate, but they are formulated within a framework that assumes the universe is a closed entity with its own laws. RTS proposes a shift of perspective: our universe is not a closed entity, but a stable slice of a larger Network.

2. FINITE OR INFINITE?

In RTS, this question is wrongly formulated. Our observable universe is finite — it is a stable region of the Network, with a certain density of couplings and a certain geometry. But the Network that contains it may be infinite in possibilities: there are countless reality slices (C_α), each with its own laws and its own degree of stability.

Thus, the RTS answer is:

- Our universe is finite in manifestation, but infinite in connections.
- Infinity is not a property of space, but a property of the Network's possibilities.

3. PENROSE'S CYCLES – PHASE TRANSITIONS

Penrose proposes an infinite cycle of aeons, in which each universe transforms into the next after all mass decays. In RTS, this process is reinterpreted as a phase transition of the Network.

When the Network reaches a critical density, the couplings reconfigure, and the old reality slice dissolves into a state of pure potential. From this state, a new slice can emerge — not through a repetition of time, but through a reconfiguration of couplings.

Penrose's cycles are, in RTS, cycles of the Network, not of time. Time is a local property of each slice, not a global dimension.

4. ENERGY – A LOCAL PROPERTY

In general relativity, the total energy of the universe is not well-defined. In RTS, this is a natural consequence of the fact that energy is an emergent property of nodal couplings.

Energy is not globally conserved because the Network is not a closed system. It is open — it can change the configuration of couplings, transfer energy between reality slices, and store energy in the subquantum wall.

Thus, RTS states: "Energy is not a universal constant. It is a local property, depending on the configuration of the Network in each slice."

5. CONCLUSION – AN OPEN PERSPECTIVE

RTS does not reject the questions of contemporary cosmology, but reformulates them. Instead of asking: "Is the universe finite or infinite?", we ask: "How does the Network manifest in this slice?"

And the answer is: the Network is infinite in possibilities, but each slice is a finite and stable manifestation of them.

WHAT'S NEW IN SCIENCE? – RTS26

"Infinity and Cycles: An RTS Perspective on Contemporary Cosmology"

This episode offers a reinterpretation of the fundamental questions of cosmology through the lens of RTS:

1. **Infinity is reformulated** — not as a property of space, but as a property of the Network's possibilities.
2. **Penrose's cycles** — reinterpreted as phase transitions of the Network, not as temporal repetitions.
3. **Energy** — defined as a local property of couplings, not as a universal constant.
4. **The universe** — seen as a stable slice of a larger Network, not as a closed entity.

In one sentence:

RTS26 transforms the questions of contemporary cosmology about infinity, cycles, and energy into an open perspective, in which the Network is infinite in possibilities, and each slice is a finite and stable manifestation.